

Course Code: CS3432

Name of the Programme: B.Tech

Name of the Course: Ethical Hacking Essentials

Semester: VI

Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
2:0:2	30+0+30	60

Course Objectives:

- a) To familiarize cybercrime terminologies and perspectives.
- b) Identify and Classify Cyber Threats and Crimes and measures to enable security.
- c) Explain the risk assessment process and common risk mitigation strategies used in information security.
- d) Gain Hands-on Experience with Cybersecurity Tools.

Course outline (Syllabus of the course)

Syllabus:

Module – 1 Overview of Cyber Security and Ethical Hacking	No. of hours
	06
The cyber kill chain methodology, understanding tactics, techniques, and procedures (TTPs), overview of indicators of compromise (IoCs), overview of hacking concepts and hacker classes, different phases of hacking cycle, ethical hacking concepts and its scope, ethical hacking tools, threats and threat sources, malware and components of malware, malware countermeasures, vulnerability classification, vulnerability assessment, vulnerability management life cycle, vulnerability assessment tools and exploitation.	

Module - 2:	No. of hours
	06
Password Security, Authentication, and Countermeasures Password cracking and password complexity, Microsoft authentication, various types of password attacks, password cracking tools, countermeasures against password attacks, social engineering techniques, insider threats, identity theft, social engineering counter measures, identity theft countermeasures.	

Module – 3	No. of hours
	06

Network and Web Application Security

Packet sniffing and types of sniffing, various sniffing techniques and tools, different sniffing countermeasures, types of DoS and DDoS -attacks, tools and counter measures, session hijacking and its types, different session hijacking tools and countermeasures, web server concepts and attacks, web server attack tools and countermeasures, web application architecture and vulnerability stack, different web application threats and attacks, web application tools, different types of SQL injection attacks, SQL injection tools, SQL injection attacks countermeasures.

Module – 4	No. of hours
	06

Wireless and Mobile Security

Overview of wireless terminology, wireless encryption algorithms, wireless network specific attack techniques, different wireless attack tools, Bluetooth attack techniques, various wireless attack countermeasures, different wireless security tools, anatomy of mobile attack, mobile platform - attack vectors, vulnerabilities, device management, security guidelines and security tools.

Module - 5	No. of hours
	06

IoT, OT, and Cloud Security

IoT - attacks and tools, OT-attacks and tools, IoT and OT attack countermeasures and security tools, cloud computing concepts, container technology, cloud computing threats, cloud attacks and tools, cloud attack countermeasures, cloud computing security tools, penetration testing fundamentals.

Laboratory Experiment List

1	Cyber Kill Chain Simulation Practice mapping a cyber attack using the cyber kill chain stages to understand attacker tactics and techniques.
2	Using Ethical Hacking Tools Hands-on with popular ethical hacking tools (e.g., Nmap, Metasploit) to scan and identify vulnerabilities in a test environment.

3	Malware Analysis Basics Analyze a sample malware in a controlled sandbox environment to identify components and behavior.
4	Vulnerability Assessment and Exploitation Perform vulnerability scanning using tools like Nessus or OpenVAS and exploit a low-risk vulnerability in a lab setup.
5	Password Cracking and Complexity Testing Use password cracking tools (e.g., John the Ripper, Hashcat) to test password strength and understand common attacks.
6	Social Engineering Simulation Conduct a controlled social engineering test (e.g., phishing simulation) and learn countermeasures.
7	Packet Sniffing and Traffic Analysis Capture and analyze network packets using Wireshark to identify sniffing attacks and sensitive data.
8	DoS/DDoS Attack Simulation and Defense Simulate a DoS attack on a local server and apply countermeasures like rate limiting and firewall rules.
9	Session Hijacking Attack and Prevention Perform a session hijacking attack in a controlled environment and implement session management countermeasures.
10	SQL Injection Testing and Mitigation Test for SQL injection vulnerabilities on a vulnerable web app and apply coding best practices to fix them.
11	Wireless Network Attacks and Defense Conduct wireless network attacks like deauthentication and rogue access point attacks, then apply encryption and security tools.
12	IoT and Cloud Security Hands-on Explore IoT device vulnerabilities and perform basic penetration testing on a cloud platform, learning tools and defenses.

Course outcomes	
a)	Explain the core concepts of ethical hacking and common computer-based vulnerabilities to demonstrate foundational knowledge of security risks and mitigation strategies.
b)	Evaluate password cracking, social engineering techniques, and their countermeasures to strengthen network and system security.

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| c) Examine hacking techniques and tools used across mobile, wireless, network-level, IoT, OT, web applications, and cloud environments to assess potential threats and defense mechanisms. |
| d) Apply ethical hacking tools to identify and report system vulnerabilities through hands-on testing and simulation. |

Text books	
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| 1 | Michael T. Simpson, Kent Backman, and James E. Corley, Hands-On Ethical Hacking and Network Defense, Course Technology, Delmar Cengage Learning, 2011, ISBN: 9781133169031 |
| 2 | The Basics of Hacking and Penetration Testing - Patrick Engebretson, SYNGRESS, Elsevier, 2013, ISBN: 9780124116443 |

Reference Books	
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| 1 | The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws, Dafydd Stuttard and Marcus Pinto, 2011, ISBN: 9781118079614. |
| 2 | https://eccouncil.org/academia/course/ethical-hacking-essentials?logged=false |